WAD - IMP PYQ Analysis

WEB APPLICATION DEVELOPMENT | BE05000281 | Sem 5 | GTU

New Syllabus 2024-25 | Old Papers Analyzed: 13 (S2016 – W2025)

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Chapter Reference

New Ch.	Covers
Ch. 1	Introduction to Web Tech: Client-Server, Static vs Dynamic, WWW, Banking/Finance role
Ch. 2	HTML & CSS Fundamentals: Tags, Forms, Flexbox, Responsive, Bootstrap basics
Ch. 3	JavaScript Fundamentals: DOM, Events, Async, Fetch API, Local Storage
Ch. 4	APIs & HTTP Communication: CRUD, JSON/XML, Status Codes
Ch. 5	Backend with Node.js: Express, REST API, DB Connectivity, Postman *(Zero PYQ - Blind Spot)*
Ch. 6	Frontend with React.js: JSX, Hooks, Props/State, CORS *(Zero PYQ - Blind Spot)*
Ch. 7	Deployment & Modern Web: Hosting, SEO, CAPTCHA, Cloud *(Zero PYQ - Blind Spot)*

M.IMP Questions - High → Low Priority

> **Priority Rule:** Freq ≥ 6 years OR foundational GTU-Evergreen topic : (HTTP/CSS/DOM) → **M.IMP**

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1. Regex-Based Login Form Validation (Username/Password/Email/Mobile)

(Ch. 2 + Ch. 3)	Freq: **7x**	Max Marks: **7**	`Diag-Code: 7`

2. HTTP Request-Response Mechanism + Methods/Headers/Status Codes

(Ch. 4)	Freq: **6x**	Max Marks: **7**	`Theory: 6`

3. CSS Types: Inline vs Embedded vs External + Class vs ID Selector

(Ch. 2)	Freq: **6x**	Max Marks: **7 / 4**	`Theory: 6`

4. DOM Manipulation + Event Handling (Mouseover, Key Events, Popups)

(Ch. 3)	Freq: **6x**	Max Marks: **7**	`Theory: 2` `Diag-Code: 4`

5. JS Logic Programs: Prime / Fibonacci / Odd-Even / Pattern Printing

(Ch. 3)	Freq: **6x**	Max Marks: **7**	`Num: 6`

6. HTML Table with Rowspan/Colspan (Reproduce-the-Table Type)

(Ch. 2)	Freq: **5x**	Max Marks: **7**	`Diag-Code: 5`

7. CSS Positioning (Relative/Absolute/Z-Index) + Flexbox Layout

(Ch. 2)	Freq: **4x**	Max Marks: **4**	`Theory: 4`

8. Client-Side Scripting vs Server-Side Scripting

(Ch. 1 / Ch. 3)	Freq: **8x**	Max Marks: **3-4**	`Theory: 8`

9. Pure HTML Form Controls & Attributes (Select, Input, Textarea, Radio, Checkbox)

(Ch. 2)	Freq: **5x**	Max Marks: **7**	`Diag-Code: 5`

IMP Questions - High → Low Priority

> **Priority Rule:** Freq 2-3 years OR recurring 3/4-mark short answer → **IMP**

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1. Callbacks in JavaScript

`(Ch. 3)` | Freq: 3x | Max Marks: **3-4** | `Theory: 3`

2. JSON - Definition, Syntax, Use Cases

`(Ch. 4)` | Freq: 3x | Max Marks: **3-4** | `Def: 3`

3. GET vs POST Methods

`(Ch. 4)` | Freq: 3x | Max Marks: **3-4** | `Theory: 3`

4. Synchronous vs Asynchronous Web Programming

`(Ch. 3)` | Freq: 3x | Max Marks: **4** | `Theory: 3`

5. HTML Meta Tags (SEO, Caching, Expiration)

`(Ch. 2 / Ch. 7)` | Freq: 3x | Max Marks: **3-4** | `Def: 3`

6. CORS - Definition + Basic Implementation

`(Ch. 6)` | Freq: **2x** | Max Marks: **3-4** | `Theory: 1` `Diag-Code: 1`

7. Website Architecture / Client-Server Diagram

`(Ch. 1)` | Freq: **2x** | Max Marks: **3-4** | `Diag: 2`

8. Popup Boxes in JS (Alert/Prompt/Confirm)

`(Ch. 3)` | Freq: **2x** | Max Marks: **3** | `Def: 2`

9. Bootstrap Basics & Grid System

`(Ch. 2)` | Freq: **4x** | Max Marks: **3-4** | `Theory/Code: 4`

New Syllabus Blind Spots - High → Low Priority

> ⚠️ **Zero PYQ history.** Old papers can't cover these - study from official docs, not past papers.

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1. Node.js + Express.js (REST API Dev, Routing, Middleware Execution Order, Event Loop)

`(Ch. 5)` | Freq: 0x - *Flagship PHP replacement.*
GTU tests new additions hard in Year 1 to prove syllabus relevance. → Likely: **Theory + Code**

2. React.js (Real DOM vs Virtual DOM, JSX, Props/State, Component Lifecycle via useEffect)

`(Ch. 6)` | Freq: 0x - *Biggest single addition, 7 dedicated hours.*
Guaranteed 7-marker debut. → Likely: **Theory + Code Snippet**

3. Fetch API + Promises (Async Patterns)

`(Ch. 3/4)` | Freq: 0x - *Direct evolution of old AJAX-style questions, same skeleton.*
→ Likely: **Code**

4. Query Parameters vs. Request Body & HTTP Verbs (GET, POST, PUT, DELETE)

`(Ch. 4)` | Freq: 0x - *Direct syllabus mapping for CRUD operations over HTTP.* → Likely: **Theory/Code**

5. Postman API Testing

`(Ch. 5)` | Freq: 0x - *Named explicitly in syllabus as a tool.*
Easy 3-marker definitional question. → Likely: **Definition**

6. MongoDB/MySQL Connectivity via Node.js (CRUD)

`(Ch. 5)` | Freq: 0x - *Replaces old "PHP+MySQL CRUD" question, same shape just Node-flavored.* → Likely: **Code**

7. Authentication/Authorization Overview

`(Ch. 5)` | Freq: 0x - *New buzzword-topic, likely shallow intro-level definitional ask.* → Likely: **Theory**

8. Browser Local Storage

`(Ch. 3)` | Freq: 0x - *Explicitly named, easy pairing with DOM/JS questions.* → Likely: **Theory/Code**

9. Deployment & Cloud Concepts (DNS Domain Management, Hosting, CAPTCHA bot mitigation)

`(Ch. 7)` | Freq: 0x - *Smallest chapter (3 hrs) but GTU tests every chapter at least once.* → Likely: **Theory**

10. Axios/Fetch for React-Backend Integration

`(Ch. 6)` | Freq: 0x - *Capstone-style question combining React+API, likely the "mini project" 7-marker.* → Likely: **Code**

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